BUYPASS® Platform
ATL105 Specification Updates
for Dynamic Card Table Processing

Version: 2014-1

June 27, 2014

This information is confidential and proprietary of First Data Corporation.
Reproduction without the expressed written consent of First Data Corporation is prohibited.

Copyright

© 2014 First Data Corporation

All rights reserved. All information contained herein is confidential and proprietary to First Data Corporation. It shall not be disclosed, duplicated, or used in part or in whole, for any purpose without prior written consent from First Data Corporation. All trademarks, service marks and trade names referenced herein are the property of their respective owners.

Disclaimer

The Dynamic Card Table Addendum is a new functionality for the BUYPASS® Platform ATL105 Message Format Specifications and is subject to change. This document should be used in conjunction with BUYPASS® Platform ATL105 Message Format Specifications.

Please subscribe to this Dynamic Card Table Addendum by checking the subscribe button on the First Data specification website prior to download, so notification will be emailed when new versions are available. Please send an email to certsupport@firstdata.com for additional information.



This information is confidential and proprietary of First Data Corporation.
Reproduction without the expressed written consent of First Data Corporation is prohibited.

Revision History

This is a new document.

The following table lists added, updated, and deleted pages:

Note: Red text and change bars in the page margin indicate updated/added text. Deleted text is indicated only in the following table. Blue text indicates a link to another location in the text.

Changes Made to Ver. No. 2014-1 Released June 27, 2014		
Topic	**2014-1 Page No.**	**Summary of Change**
General	N/A	This is the initial version.

This information is confidential and proprietary of First Data Corporation.
Reproduction without the expressed written consent of First Data Corporation is prohibited.

Table of Contents

1 Introduction 1
1.1 Purpose of Dynamic Card Table 1
1.2 Scope of the Document 1
1.3 References 1
2 Application Architecture 3
2.1 Processing Overview 3
2.1.1 Dynamic Card Tables Overview 3
2.1.2 Dynamic Card Table Requirements from User Standpoint 4
2.1.3 Dynamic Card Table Impacts on the Host and POS Interface 5
2.1.4 How does the Host and Device Interact 7
2.2 Card Tables 8
2.2.1 BIN Table 9
2.2.2 RULES Table 11
2.2.3 SAF Table 28
2.2.4 RESTRICTIONS Table 30
2.2.5 PRODUCT CODE Table 32
2.2.6 PROMPT Table 34
2.3 Data Format in Proprietary Data Load Request 39
2.4 Host Databases for Managing the Process 39
2.4.1 Card Table Master 39
2.4.2 Merchant Card Table 42
2.4.3 BANKODFI 43
2.5 User Interface 44
2.5.1 Reports 44
2.5.2 Screens 44
2.5.3 Transaction Terminals 44
2.6 Database 44
2.7 Recovery 44
2.8 Security 45
2.9 Diagnostics 45
2.10 Performance 45
3 Environmental Architecture 46
3.1 Hardware Platform 46

This information is confidential and proprietary of First Data Corporation.
Reproduction without the expressed written consent of First Data Corporation is prohibited.

3.2 Software Platform 46
4 Departmental Impacts 47
4.1 QA/Certification 48
4.2 Operations 48
4.3 Production Support 48
4.4 Communications 48
4.5 Technical Services 48
4.6 Billing/Balancing 48
4.7 Customer Service 48
4.8 Data Entry 48
4.9 Marketing Support 49
4.10 Depot/Maintenance 49
4.11 Accounting 49
5 Customer Impacts 50
5.1 Merchant 50
5.2 End Processor 50
5.3 Financial/Proprietary Institutions 50
5.4 Authorizers/Issuers 50

This information is confidential and proprietary of First Data Corporation.
Reproduction without the expressed written consent of First Data Corporation is prohibited.

1 Introduction

This document provides information about the requirements for implementing Dynamic Card Table processing involving the BUYPASS Platform (BUYPASS).

The primary audience of this document includes the merchants, vendors, programmers, and analysts implementing support for Dynamic Card Table processing services.

1.1 Purpose of Dynamic Card Table

The purpose of Dynamic Card Table is to provide a table driven process. Following are the advantages of the Dynamic Card Table:

- Limit the impacts to the point-of-service (POS) as it relates to the processing characteristics of a specific method of payment**.**
- Replace static values configured into the POS with customizable data configured at the BUYPASS host and downloaded to the device**.**
- Reduce or eliminate the impact of future changes, such as, the need to define a new Card Type and its capabilities, or change the characteristics of how a payment method is processed based on changing business requirements**.**

1.2 Scope of the Document

This document outlines all areas of impact based on the current definition of the Dynamic Card Table functionality. This document describes each of these areas in detail in order to define the impacts of the required changes and the resulting functionality to be delivered.

1.3 References

This document draws on information obtained during various workshops held between BUYPASS/First Data and POS Vendors. Additional input was gathered from workshops and technical reviews with customers that would potentially implement this functionality in its BETA release.

This document contains the following chapters and appendices:

1. **Introduction.** This chapter describes the purpose, scope, references, and target audience of this document. It also provides a list and describes each chapter and appendices present in the document.
2. **Application Architecture.** This chapter describes the dynamic card table processing overview, six categories of dynamic card tables, data format in Proprietary Data Load Request, host databases for managing the process, user interface, database, recovery, security, diagnostics, and performance.
3. **Environmental Architecture.** This chapter describes the impacts of dynamic card table processing on hardware and software platforms.

This information is confidential and proprietary of First Data Corporation.
Reproduction without the expressed written consent of First Data Corporation is prohibited.

4. **Departmental Impacts.** This chapter describes the impacts of dynamic card table processing on various departments like QA/Certification, Operations, Production Support, Communications, Technical Services, Billing/Balancing, Customer Service, Data Entry, Marketing Support, Depot/Maintenance, and Accounting.
5. **Customer Impacts.** This chapter describes the impacts of dynamic card table processing on various customers like merchant, end processor, financial/proprietary institutions, and authorizers/issuers.

This information is confidential and proprietary of First Data Corporation.
Reproduction without the expressed written consent of First Data Corporation is prohibited.

2 Application Architecture

This chapter discusses Dynamic Card Table impacts on the following application architecture:

- Processing Overview
- Card Tables
- Data Format in Proprietary Data Load Request
- Host Databases for Managing the Process
- User Interface
- Database
- Recovery
- Security
- Diagnostics
- Performance

2.1 Processing Overview

2.1.1 Dynamic Card Tables Overview

The Dynamic Card Table feature consists of 6 unique supporting tables, each of which establishes processing characteristics for every method of payment supported. Each individual table defines a different set of parameters which, when all combined, provide the complete processing requirements for a payment method. For example, the card table defines the expected Track I and Track II characteristics based on the method of payment as determined by a recognized BIN. It also includes, but is not limited to information, such as, product code limitations, certain prompting that should occur, and the ability to support the indicated card type in Store and Forward mode. The 6 supporting tables are defined to be the following:

- BIN table
- RULES table
- RESTRICTIONS table
- SAF table
- PROMPT table
- PRODUCT table

Each of the supporting tables is assigned its own unique version number. Versioning allows configuring devices within the network. This configuration allows the devices to process differently by assigning the devices a different version of any individual table(s).

From a hierarchy viewpoint within the BUYPASS host, a Card Table Master Database is defined. This database contains records defining the specific version of each supporting table that comprises a Master ID. For example, Master ID 1 would be comprised of BIN table 1, RULES table 5, RESTRICTIONS table 4, SAF table 3, PRODUCT table 5, and PROMPT table 5.

First Data.

This information is confidential and proprietary of First Data Corporation.
Reproduction without the expressed written consent of First Data Corporation is prohibited.

A change to a supporting table is facilitated by creating a new version of the affected table. For example, if a new PRODUCT characteristic was desired, the PRODUCT table would be updated to produce version 6. Depending on the extent of a change, updates would be noted in any of the following ways:

- Update the Master ID record to reflect it now uses the new version of the PRODUCT table, or an entirely new Master ID record would be created. For example, the BUYPASS host would allow the customer to update Master ID 1 to define its PRODUCT table as version 6 instead of the original version 5.
- Request a new Master ID 2 record that mirrors Master ID 1 with the exception of the version of the PRODUCT table. Once a Master ID is defined by identifying the version of the 6 supporting tables, it is recorded in the Card Table Master database. The Card Table Master database also contains an identifier as to the Customer/Corporate Id. This identifier is be used to distinguish between the Card Table Master for Customer A from that for Customer B.

2.1.2 Dynamic Card Table Requirements from User Standpoint

As part of the Dynamic Card Table functionality there are some basic requirements based on customer input. Following are the requirements which should be included in order to support the Dynamic Card Table functionality:

- The BUYPASS host must be capable of initiating a download of card table data. This will be managed during the End of Day Totals processing. To support this requirement, a new totals type, Totals with Proprietary Data Load Segment, has been added in ATL105 Message Format Specification. This is to support the device identifying its current card table version information so that the BUYPASS host can recognize updates that need to occur.

- The BUYPASS host must be capable of providing updates to the device for an individual card table rather than forcing complete reload of all 6 card table datasets. A new field called “Version” has been included in the new Totals request to support this requirement. The version information contains a reference to a Master Identifier and the specific versions of the component tables. The process then allows the BUYPASS host to indicate that a specific supporting table is being updated and provides that data independently of the complete card table data.

- The BUYPASS host must be capable of Line Item downloading to support simple changes. In considering this aspect of processing, there were several obstacles to a clean solution. For example, some of the supporting tables allow for duplicate ‘key’ indicators. In that situation the POS Device would not be able to distinguish an update to an existing record versus an addition of a new record. As a result the ability to perform Line Item updates is out of scope. To support this requirement BUYPASS host is allowed to support a single table update or refresh a single table update.

- The BUYPASS host must support Multiple Table Versions. This is the ability of the BUYPASS host to recognize some locations may need to have different processing characteristics than other locations. The configuration databases that have been designed, allow for multiple versions of the supporting tables to be grouped under a Master ID, with multiple Master IDs. A second database allows Merchants to be assigned to specific Master ID records, which in turn supports customer versions by location. The Merchant configuration database also supports the concept of a Global record. This allows a single record to be used to reference a default Master ID which

This information is confidential and proprietary of First Data Corporation.
Reproduction without the expressed written consent of First Data Corporation is prohibited.

alleviates the requirement for all Merchants to be defined in the database. Only exceptions to the default would need to be added. This will simplify the setup and management of the configuration.

- The BUYPASS host must support Background Downloading and Automated Site Implementation. These requirements state that the download process cannot interfere with normal financial transaction processing and cannot impact the customer. Additionally, once the load is complete the site system must implement the updates without intervention and prior to business day close. The mechanism for the BUYPASS host to proactively send updates is tied directly to the End of Day Totals processing. Therefore from the perspective of the BUYPASS host it is part of the business day close process. Once the data is provided to the site system it is out of the control of the host process. This then becomes a requirement of the site POS system. Managing the download through site system is out of scope for BUYPASS host. The BUYPASS host load process is a unique request and reply process. The data exchange contains the necessary information for the BUYPASS host to engage in a multi-block message and retain context between exchanges. The BUYPASS host does not monitor if financial transactions are interspersed between requests for the next load block. These two functions, transaction processing and data download, are handled by separate modules of the BUYPASS host and therefore may occur in parallel.
- The BUYPASS host must support Proprietary Data Load. Although not a requirement, the new Proprietary Data Load request message is a standalone request. This means that the request for a complete table load, or a supporting table load could be initiated by the device.
- The BUYPASS host specials must override the dynamic card tables. For example, if the BIN for Visa is in the card table but the host load indicates Visa is not accepted, the host overrides the definition in the card table. This requirement applies to card types that are downloaded, AVS/ZIP prompting, and SAF blocking.

2.1.3 Dynamic Card Table Impacts on the Host and POS Interface

Currently the BUYPASS host checks for existing data that may need to be sent to a device during the Totals Request process involving Data Segment No.105 (Totals Data Segment). For example, in the current implementation if there was Electronic Mail for a location, the device would be alerted to request that information as part of this process.

The Dynamic Card Table is being managed as part of the End of Day Totals processing. This involves a new request type - Totals with Dynamic Card Table Data. This new request type requires the POS to replace use of the existing Data Segment No.105 (Totals Data Segment) with the new Data Segment No. 119 (Totals with Proprietary Data Load Segment). The new segment provides a new Data Element so that the device can identify its current Card Table version data. This mechanism facilitates the host in recognizing whether an update is required to the device configuration.

The ATL105 Specification has also been updated with a new Data Segment No. 118 (Proprietary Data Load Segment). This is the segment which is used to facilitate the data exchange between the Box and the BUYPASS host in order to provide the card data to the device. This message allows the host to indicate the following:

- New Card Table Version that is being sent
- Specific supporting table being updated in a current request/reply
- A multi-message exchange for larger updates is being controlled

This information is confidential and proprietary of First Data Corporation.
Reproduction without the expressed written consent of First Data Corporation is prohibited.

The POS Device saves the Card Table Version identifier. This value is returned as part of the Totals with Proprietary Data Load Segment. The POS Device also accepts the multi-message process of a new table(s) load. Once the POS Device has received notification that the load is complete, it must manage the process of actually replacing the tables it is currently using with the newly updated tables from the load. There is no requirement from the host on the timing of this process. If the POS Device has transactions outstanding to the host when the load function is complete, it has discretion on delaying the actual table activation.

This information is confidential and proprietary of First Data Corporation.
Reproduction without the expressed written consent of First Data Corporation is prohibited.

2.1.4 How does the Host and Device Interact

On initial installation, a POS device has a defined series of load functions that occur. These loads provide the device with certain information that allows it to process on the BUYPASS host. As part of this process, the new Dynamic Card Table data is loaded. POS Device uses this data to implement some soft functionality that can be changed as needed.

A Version Identifier is a part of the Dynamic Card Table data load. The Version Identifier comprises the Card Table Master version as well as the version indicators for the 6 supporting tables. The POS device saves this value and uses the value as part of the End of Day Totals processing. This facilitates the POS Device to relay its current Card Table configuration. The BUYPASS host uses this data for the following two purposes:

1. Compare the current Card Table Master version part of the tag against the Merchant configuration record. This allows the host to verify that the device is using the correct Card Table. If an update has been made to the Card Table Master version, the host would then initiate a load to the device to replace its card table data.
2. Obtain the current versions of the 6 supporting tables. To obtain the current versions, host verifies whether the version of the Card Table Master is correct. If the version is found to be correct, host retrieves that record from the Card Table Master database. That data is then compared with the version identifier from the device. This allows the host to determine if a supporting table has been updated. If there are one or more changes, the host would initiate a load to the device to replace the affected supporting tables.

These functionalities of BUYPASS host allow some changes to how the card processing is to be handled without requiring software changes to the POS devices. It also provides flexibility for the customer to configure certain characteristics and restrictions and control it by location. The host monitors the versions at the location and updates the versions, as necessary.

A customer can load and test a configuration change to a subset of locations. For this, a customer implements a new Card Table Master version. Once the processing has been validated, the Card Table Master version can be updated to reflect the updated versions of the affected supporting tables. This results in controlled installation of new changes without the risk of unexpected impacts.

First Data.

This information is confidential and proprietary of First Data Corporation.
Reproduction without the expressed written consent of First Data Corporation is prohibited.

2.2 Card Tables

The dynamic card table functionality defines 6 new tables to provide the desired function. Each of these tables addresses a different functional area of card processing. These tables are linked together via an identifier within its parent table.

A version is tracked for each table set in order to support updates to the tables. This process of version tracking allows an update to be in a 'beta' test mode or in a controlled use environment, while other devices function on a prior version of that table.

The Card Tables exist as individual databases and the naming structure is such that the version is included. This facilitates easier management and identification of the various components and their versions.

The 6 tables are as follows:

- BIN table
- RULES table
- RESTRICTIONS table
- SAF table
- PROMPT table
- PRODUCT table

Each of these tables is defined in the following sub sections. The sub sections describes in detail the format and use of each of these tables. Additionally the link from the parent to a particular table is also defined.

Note: All the 6 tables include a column called IN Load File. This column indicates which columns from the BUYPASS host files will actually be sent to the device during a Dynamic Card Table Load process. For example, the columns labeled DESCRIPTION and CORPORATE ID will never be in the data sent to the device. The extra columns, if any, are intended for internal BUYPASS definition and use.

First Data.

This information is confidential and proprietary of First Data Corporation.
Reproduction without the expressed written consent of First Data Corporation is prohibited.

2.2.1 BIN Table

The BIN table defines the card type based on a Begin and End range for the account number. The RULES KEY defines the values from the RULES table that apply to the indicated BIN/Card type.

Dynamic Card Table
BIN table

Field Name	Max Field Length	Field Type	Values	Description	Key Column	IN Load File
Description	30	AN		Identifies the entry description.		
Corporate ID	10	AN		Identifies the customer used to distinguish the users of this entry. For example: FIRSTDATA	Y	
Version	5	N		Identifies the records that are part of a specific version.	Y	
Sequence	9	N		Identifies the Sequence Number to control ordering of BINS – Using Begin Range does not order as required in all cases for some Users.	Y	
Mode	1	AN		Identifies the Mode of Operation in which the transaction was initiated. Default will be C. **Value** **Description** C Credit card, swiped or keyed S Speedpass device reader D Customer selected Debit	Y	Y

This information is confidential and proprietary of First Data Corporation.
Reproduction without the expressed written consent of First Data Corporation is prohibited.

Dynamic Card Table BIN table						
Field Name	**Max Field Length**	**Field Type**	**Values**	**Description**	**Key Column**	**IN Load File**
BEGIN Range	12	AN		Defines the beginning BIN range value – left justified and padded with spaces.	Y	Y
END Range	12	AN		Defines the ending BIN range value – left justified and padded with spaces.	Y	Y
RULES Key	4	AN		Identifies the index into the RULES table.		Y
Timestamp	12	N		Identifies the CCYYMMDDHHMM of the last update to this entry.		

As an example, assume the following content to an entry in this table:

Description: VISA FLEET
Corporate ID: FIRSTDATA
Version: 00001
Sequence: 000000001
Mode: C
Begin Range: 448460
End Range: 448611
Rules Key: VIFT
Timestamp: 201201261000

The data sent to the POS/Box would appear as follows:

C,448460,448611,VIFT\

This information is confidential and proprietary of First Data Corporation.
Reproduction without the expressed written consent of First Data Corporation is prohibited.

2.2.2 RULES Table

The RULES table defines the parameters associated with the indicated payment type. This table also contains indexes to the other related tables.

Dynamic Card Table
RULES table

Field Name	Max Field Length	Field Type	Values	Description	Key Column	IN Load File
Description	30	AN		Identifies the entry description.		
Corporate ID	10	AN		Identifies the customer used to distinguish the users of this entry. For example: FIRSTDATA	Y	
Version	5	N		Identifies the records that are part of a specific version.	Y	
RULES Key	4	AN		Identifies the Key for accessing the BIN table used to retrieve applicable records.	Y	Y
Source	1	AN		Indicates how the account information was physically presented. Default will be M. **Value** **Description** M Mag stripe, either Track 2 or Track 1 K Keyed S Speedpass device reader	Y	Y

This information is confidential and proprietary of First Data Corporation.
Reproduction without the expressed written consent of First Data Corporation is prohibited.

Dynamic Card Table
RULES table

Field Name	Max Field Length	Field Type	Values	Description	Key Column	IN Load File
Track 2 Match	30	AN		Defines the parsing of the mag stripe data and any actions to take, based on the values present in indicated positions. See the detail following this table for how this field is constructed.	Y	Y
Track 1 Match	30	AN		Defines the parsing of the mag stripe data and any actions to take, based on the values present in indicated positions. See the detail following this table for how this field is constructed.		Y
Track Preference	1	N		Determines the preferred track to attempt to match in the Rules Table, and send in the request message. If the preference doesn't match, attempt to match the alternate track. Default is 2. **Value** **Description** 1 Track 1 is preferred in a request 2 Track 2 is preferred in a request		Y

This information is confidential and proprietary of First Data Corporation.
Reproduction without the expressed written consent of First Data Corporation is prohibited.

Dynamic Card Table RULES table						
Field Name	**Max Field Length**	**Field Type**	**Values**	**Description**	**Key Column**	**IN Load File**
Receipt Text	12	AN		Identifies the card type description to be printed on Receipts and any detail reports. Can be upper and lower case text.		Y
Debit Prompt	1	N		Indicates to prompt customer select processing for cards that could be Credit or Debit. Default is 1. **Value** **Description** 0 Do not prompt for Debit or Credit 1 Prompt customer to select 2 Debit Encouragement: customer is requested to "ENTER PIN" but can press "CANCEL" to proceed as Credit.		Y

This information is confidential and proprietary of First Data Corporation.
Reproduction without the expressed written consent of First Data Corporation is prohibited.

Dynamic Card Table
RULES table

Field Name	Max Field Length	Field Type	Values	Description	Key Column	IN Load File
HOST Route	1	N		Defines to the device if this BIN should be routed to host for authorization or use an alternate authorization process. For example Gift Card Mall or Loyalty applications. Default is 1. **Value** **Description** 0 Do not route to BUYPASS host for Authorization 1 Route to BUYPASS host for Authorization		Y
Low Fee Card	1	N		Defines card eligible for Local Discounting by merchant based on associated processing fees. Default is 0. **Value** **Description** 0 Card is NOT a low fee card 1 Card is a low fee card		Y

This information is confidential and proprietary of First Data Corporation.
Reproduction without the expressed written consent of First Data Corporation is prohibited.

Dynamic Card Table
RULES table

Field Name	Max Field Length	Field Type	Values	Description	Key Column	IN Load File
Block Pre-Auth	1	N		Identifies whether the card supports Pre-Auth transactions. Default is 0. **Value** **Description** 0 Card supports Pre-Auth transactions 1 Card does not support a Pre-Auth transaction. Transactions from an Outdoor device are not supported for this Card Type. Indoor transactions are supported for Pre-Pay as a Purchase only. If the final charge is less than the Pre-Pay amount, the system must issue a refund for the difference. Currently this is limited to EBT only		Y

This information is confidential and proprietary of First Data Corporation.
Reproduction without the expressed written consent of First Data Corporation is prohibited.

Dynamic Card Table
RULES table

Field Name	Max Field Length	Field Type	Values	Description	Key Column	IN Load File
Retain Track Data	1	N		Identifies the Track Data information for follow-on transactions, such as Completions and Reversals. Therefore the device must retain this information for subsequent transactions. Default is 0. **Value** **Description** 0 No need to retain Track data at the device 1 Device must retain Track data for follow-on transactions		Y
Decline Receipt	1	N		Indicates whether this card type requires the printing of a receipt on a declined transaction. Default is 0. **Value** **Description** 0 No receipt printing is required 1 Receipt must be printed on a declined transaction		Y

This information is confidential and proprietary of First Data Corporation.
Reproduction without the expressed written consent of First Data Corporation is prohibited.

Dynamic Card Table
RULES table

Field Name	Max Field Length	Field Type	Values	Description	Key Column	IN Load File
DISCOUNT STACKING	1	N		Defines how discounts are to be applied when multiple discounts are available for a single transaction. This includes at what point in the transaction an EPS/POS Retailer, 3rd Party, and/or host Based discount is applied and reflected. **Value** **Description** 0 No stacking allowed 1 Local discount may be applied and price rolled back at pump, 3rd party and host discounts applied after handle is hung up.		Y

This information is confidential and proprietary of First Data Corporation.
Reproduction without the expressed written consent of First Data Corporation is prohibited.

Dynamic Card Table
RULES table

Field Name	Max Field Length	Field Type	Values	Description	Key Column	IN Load File
				Value **Description** 2 Host discount may be applied and price rolled back at pump, 3rd party and local discounts applied after handle is hung up 3 Host and local discounts may be stacked and price rolled back at pump, 3rd party discount applied after handle is hung up 4 3rd party discount may be applied and price rolled back at pump, host and local discounts applied after handle is hung up.		

This information is confidential and proprietary of First Data Corporation.
Reproduction without the expressed written consent of First Data Corporation is prohibited.

Dynamic Card Table
RULES table

Field Name	Max Field Length	Field Type	Values	Description	Key Column	IN Load File
				Value **Description** 5 3rd party and local discounts may be stacked and price rolled back at pump, host discount applied after handle is hung up 6 3rd party and host discounts may be stacked and price rolled back at pump, local discount applied after handle is hung up 7 3rd party, host and local discounts may be stacked and price rolled back at the pump		
SAF Key	4	N		Identifies the Key for accessing the SAF table.		Y
Restrictions Key	4	N		Identifies the Key for accessing the RESTRICTIONS table.		Y

This information is confidential and proprietary of First Data Corporation.
Reproduction without the expressed written consent of First Data Corporation is prohibited.

Dynamic Card Table
RULES table

Field Name	Max Field Length	Field Type	Values	Description	Key Column	IN Load File
Prompt 1	2	AN		Identifies the Key for accessing PROMPT Table entry (up to 3 allowed). Default is spaces.		Y
Prompt 2	2	AN		Identifies the Key for accessing PROMPT Table entry (up to 3 allowed). Default is spaces.		Y
Prompt 3	2	AN		Identifies the Key for accessing PROMPT Table entry (up to 3 allowed). Default is spaces.		Y

This information is confidential and proprietary of First Data Corporation.
Reproduction without the expressed written consent of First Data Corporation is prohibited.

Dynamic Card Table RULES table						
Field Name	**Max Field Length**	**Field Type**	**Values**	**Description**	**Key Column**	**IN Load File**
Totals Group	3	AN		Identifies the Card Label categories. Default is CC **Value** **Description** CC Mastercard, Visa TE American Express DS Discover, Diners, JCB DB Debit FL Fleet CS Cash PR Proprietary CK Check EF EBT Food Stamp EC EBT Cash Benefit SV Stored Value ECA Telecheck ECA EWC EBT version of WIC EK EBT Child Care		Y

This information is confidential and proprietary of First Data Corporation.
Reproduction without the expressed written consent of First Data Corporation is prohibited.

Dynamic Card Table RULES table						
Field Name	**Max Field Length**	**Field Type**	**Values**	**Description**	**Key Column**	**IN Load File**
HOST Card Type	4	N		Identifies the BUYPASS Internal Card type value. This value is to be used in conjunction with Host Discounting. The Host Discount functionality identifies by BUYPASS Internal Card Type a discount to apply to a transaction before submitting for authorization. Default is 0		Y

This information is confidential and proprietary of First Data Corporation.
Reproduction without the expressed written consent of First Data Corporation is prohibited.

Dynamic Card Table
RULES table

Field Name	Max Field Length	Field Type	Values	Description	Key Column	IN Load File
Card Type PROMPT Code	3	AN		Used with Data Element No. 14 (Card Type) and/or Data Element No 78 (Prompt code). Default is 0. **Value** **Description** 001 BUYPASS Fleet 011 DBT Checking 012 DBT Saving 013 DBT ACH 020 Credit 040 Loyalty 041 Telecredit 042 Telecheck Service 045 Generic Check 446 TeleCheck ECA 050 Reserved 051 Reserved 055 Reserved 056 Generic Proprietary 057 Reserved 058 Reserved 059 Wright Express 060 Voyager 064 Reserved 070 Reseved 071 Fleet One 072 Reserved 073 EBT Food Stamps 074 EBT Cash Benefits 075 Reserved 077 Reserved		Y

First Data.
This information is confidential and proprietary of First Data Corporation.
Reproduction without the expressed written consent of First Data Corporation is prohibited.

Dynamic Card Table
RULES table

Field Name	Max Field Length	Field Type	Values	Description	Key Column	IN Load File
				Value **Description** 072 Reserved 073 EBT Food Stamps 074 EBT Cash Benefits 075 Reserved 077 Reserved 078 Phone 079 Stored 080 Mastercard Fleet 081 Fuelman/Gascard 083 Cardlock Fuel Systems, Inc Fleet 084 Reserved 085 Reserved		
Signature Required Indicator	1	AN	Y/N	Indicates whether signature is required on receipt. Default is N.		Y
Signature Required Amount	5	N		Identifies the minimum amount before signature is required.		Y
Refund allowed	1	AN	Y/N	Indicates whether card type allows Merchandise Return. Default is Y.		Y

This information is confidential and proprietary of First Data Corporation.
Reproduction without the expressed written consent of First Data Corporation is prohibited.

Dynamic Card Table
RULES table

Field Name	Max Field Length	Field Type	Values	Description	Key Column	IN Load File
CVV Prompt	2	AN		Identifies the Key for accessing PROMPT table when CVV Prompting is enabled via Special Function 174. Default is Spaces.		Y
AVS Prompt	2	AN		Identifies the Key for accessing PROMPT table when AVS Prompting is enabled via Special Function 127. Default is Spaces.		Y
Requested Amount	7	N		Identifies the amount to be used for Preauthorized Request for use with AFP's to set a default amount. If the BUYPASS host has an entry in the AFPAMT table, the host will override this value. Regardless of the Requested Amount in the RULES table or the host AFP Amount setting, the device should activate based on the Approved Amount in the reply message. This is necessary for Partial Approval Amount compliance. The format for this data is 999v99.		Y
Timestamp	12	N		Identifies the CCYYMMDDHHMM of the last update to this entry.		

This information is confidential and proprietary of First Data Corporation.
Reproduction without the expressed written consent of First Data Corporation is prohibited.

The following details how the Track I and Track II Match formatting is constructed:

[nn] = nn digits in account number, last digit is a Luhn check digit

(nn) = nn digits in account number, no check digit

YY = two (2) digit year value to be validated as unexpired

MM = two (2) digit month value to be validated as unexpired

yy = two (2) digit year value, no validation to be performed

mm = two (2) digit month value, no validation to be performed

z = variable number of characters to be ignored

X = track does not exist on mag stripe or is not valid

<nn> = skip nn characters

Any other value = constant value to be matched in that position. If a constant value falls at end of the model string, it must be the last character in the track read by the POS

As an example, assume the following content on this table:

Description:	Visa Fleet - 01
Corporate ID:	FIRSTDATA
Version:	00002
RULES Key:	VIFT
Source:	M
Receipt Text:	Visa Fleet
Track 2 Match:	[16]=YYMMz01
Track 1 Match:	[16]^z^YYMMz01
Track Preference:	0
Debit Prompt:	0
Host Route:	1
Low Fee Card:	0
Block Preauth:	0
Retain Track:	0
Decline Receipt:	0
Discount Stacking:	0
SAF Key:	0000
Restrictions Key:	0001
Prompt 1:	FM
Prompt 2:	FG

This information is confidential and proprietary of First Data Corporation.
Reproduction without the expressed written consent of First Data Corporation is prohibited.

Prompt 3:

Totals Bucket:	CC
BUYPASS Card Type:	0004
Card Type PROMPT Code:	020
Signature Required Indicator:	N
Signature Required Amount:	10000
Refund Allowed:	Y
CVV Prompt:	ZC
AVS Prompt:	ZD
Timestamp:	201201271530

The data sent to the POS/Box would appear as follows:

VIFT,M,Visa Fleet,[16]=YYMMz01,[16]^z^YYMMz01,
0,0,1,0,0,0,0,0,0,1,FM,FG,,CC,
0004,020,N,10000,Y,ZC,ZD\

This information is confidential and proprietary of First Data Corporation.
Reproduction without the expressed written consent of First Data Corporation is prohibited.

2.2.3 SAF Table

The SAF table defines certain attributes related to Store and Forward support for the indicated entry. The SAF KEY is the link from the RULES table to the SAF table. These values apply for transactions processed while the device is in SAF mode, meaning that it has no connectivity to the BUYPASS host.

Dynamic Card Table
SAF table

Field Name	Max Field Length	Field Type	Values	Description	Key Column	IN Load File
Description	30	AN		Identifies the entry description.		
Corporate ID	10	AN		Identifies the customer used to distinguish the users of this entry. For example: FIRSTDATA	Y	
Version	5	N		Identifies the records that are part of a specific version.	Y	
SAF Key	4	N		Identifies the Key for accessing the RULES table that links to a single entry in the SAF table.	Y	Y
Max Num IN	2	N		Indicates the maximum number of indoor sales allowed in a day for a given card. Default is 0.		Y
Max Amt IN	7	N		Indicates the maximum amount of indoor sales allowed in a single transaction for a given card. Format is 99999v99 with a default of 0.		Y
Max Num Out	2	N		Indicates the maximum number of outdoor/pump sales allowed in a day for a given card. Default is 0.		Y

First Data.

This information is confidential and proprietary of First Data Corporation.
Reproduction without the expressed written consent of First Data Corporation is prohibited.

Dynamic Card Table
SAF table

Field Name	Max Field Length	Field Type	Values	Description	Key Column	IN Load File
Max Amt Out	7	N		Indicates the maximum amount of outdoor/pump sales allowed in a single transaction for a given card. Format is 99999v99 with a default of 0.		Y
Timestamp	12	N		Identifies the CCYYMMDDHHMM of the last update to this entry.		

As an example, assume the following content on this table:

Description: SF NOT ALLOWED

Corporate ID: FIRSTDATA

Version: 00003

SAF Key: 0000

Max Num IN: 0

Max Amt IN: 0

Max Num Out: 0

Max Amt Out: 0

Timestamp: 201201281900

The data sent to the POS/Box would appear as follows:

0,0,0,0,0\

This information is confidential and proprietary of First Data Corporation.
Reproduction without the expressed written consent of First Data Corporation is prohibited.

2.2.4 RESTRICTIONS Table

The RESTRICTIONS table defines transaction amount limits placed on certain product codes. The RESTRICTIONS KEY is the link from the RULES table to the RESTRICTIONS table.

Dynamic Card Table
RESTRICTIONS table

Field Name	Max Field Length	Field Type	Values	Description	Key Column	IN Load File
Description	30	AN		Identifies the entry description.		
Corporate ID	10	AN		Identifies the customer used to distinguish the users of this entry. For example: FIRSTDATA	Y	
Version	5	N		Identifies the records that are part of a specific version.	Y	
RESTRICTIONS Key	4	N		Identifies the Key for accessing RULES table that links to one or more entries in the RESTRICTIONS table.	Y	Y
Restrictions Sequence Number	5	N		Identifies the Sequence Number which is incremental from 1 (one) for each duplicate Restrictions Key entry to ensure order. For example, a Product Code of "***" would appear before 00* otherwise. This ensures entries within a Key are ordered as they should be applied.	Y	

First Data.

This information is confidential and proprietary of First Data Corporation.
Reproduction without the expressed written consent of First Data Corporation is prohibited.

Dynamic Card Table
RESTRICTIONS table

Field Name	Max Field Length	Field Type	Values	Description	Key Column	IN Load File
Product Code	3	N		Identifies the Product Code to be limited. Valid values are based on the NACS Product Codes. A wildcard of "*" allows any numeric value in that position of the Product Code. Default is 0	Y	Y
Max Amt	7	N		Identifies the maximum dollar amount of a single purchase for the indicated Product Code. A value of $0.00 totally prohibits the indicated product code from use. Default is 99999v99.		Y
Timestamp	12	N		Identifies the CCYYMMDDHHMM of the last update to this entry.		

As an example, assume the following content on this table:

Description: EXCLUDE CASH BACK

Corporate ID: FIRSTDATA

Version: 00004

RESTRICTIONS Key: 0001

Restrictions Seq Num: 1

Product Code: 955

Max Amt: 0

Timestamp: 201201241100

The data sent to the POS/Box would appear as follows:

1,955,0\

This information is confidential and proprietary of First Data Corporation.
Reproduction without the expressed written consent of First Data Corporation is prohibited.

2.2.5 PRODUCT CODE Table

The Product Code table defines the valid list of Product Codes supported as well as provides a primary and secondary text value associated with this product. In most cases only a primary text value will be in use but is provided as a special request.

Dynamic Card Table
PRODUCT CODE table

Field Name	Max Field Length	Field Type	Values	Description	Key Column	IN Load File
Description	30	AN		Identifies the entry description.		
Corporate ID	10	AN		Identifies the customer used to distinguish the users of this entry. For example: FIRSTDATA	Y	
Version	5	N		Identifies the records that are part of a specific version.	Y	
PRODUCT Code	3	N		Identifies the NACS Product Code value being supported.	Y	Y
Primary Text Value	15	AN		Identifies the Primary Product Name to be used for displays/receipts. Default is “MISC”.		Y
Secondary Text Value	15	AN		Identifies the Secondary Product Name to be used for displays/receipts.		Y
Timestamp	12	N		Identifies the CCYYMMDDHHMM of the last update to this entry.		

As an example, assume the following content on this table:

Description: Cash Back

Corporate ID: FIRSTDATA

Version: 00005

PRODUCT Code: 955

This information is confidential and proprietary of First Data Corporation.
Reproduction without the expressed written consent of First Data Corporation is prohibited.

Primary Text Value: CASH BACK

Secondary Text Value: CASH BACK

Timestamp: 201201231445

The data sent to the POS/Box would appear as follows:

995,CASH BACK,CASH BACK\

Note: This could also be configured with the Secondary Text Value of spaces and the resulting data to POS/Box would appear as:

995,CASH BACK,\

Based on the described usage of this data, the absence of Secondary Text should default to matching the Primary Text. The net result to the device should be the same in both cases.

The purpose of Secondary Text for this functionality is that some customers have locations branded to more than 1 fuel retailer. The BUYPASS host will not have any way to determine which value is applicable to a location. Therefore the host is dependent on the POS to determine which text is applicable when both are present in a Product Code entry record.

This information is confidential and proprietary of First Data Corporation.
Reproduction without the expressed written consent of First Data Corporation is prohibited.

2.2.6 PROMPT Table

The PROMPT table defines certain attributes related to prompting of input or special values to the customer/cashier. The PROMPT CODE is the link from the RULES table to the PROMPT table. Additionally, the CVV Prompt and AVS Prompt in the RULES table will be applicable based on the Option Setting for Prompt for Zip Code/AVS (Special 127) and Prompt for CVV (Special 174).

Dynamic Card Table
PROMPT table

Field Name	Max Field Length	Field Type	Values	Description	Key Column	IN Load File
Description	30	AN		Identifies the entry description.		
Corporate ID	10	AN		Identifies the customer used to distinguish the users of this entry. For example: FIRSTDATA	Y	
Version	5	N		Identifies the records that are part of a specific version.	Y	
PROMPT Code	2	AN		Identifies the Key for accessing the RULES table that links to up to 3 entries in the PROMPT table.	Y	Y
Display Line 1	16	AN		Identifies the first line of display text. Default is “ENTER DATA”.		Y
Display Line 2	16	AN		Identifies the second line of display text. Default is spaces.		Y

First Data.

This information is confidential and proprietary of First Data Corporation.
Reproduction without the expressed written consent of First Data Corporation is prohibited.

Dynamic Card Table
PROMPT table

Field Name	Max Field Length	Field Type	Values	Description	Key Column	IN Load File
Data Type	1	AN		Specifies the type of input data this prompt accepts. Default is N. **Value** **Description** # Numeric up to 20 digits right justified and zero filled N Numeric up to 20 digits left justified and space filled $ Up to 15 digit dollar amount as 9(13)V99 A Alphanumeric up to 20 characters left justified and space filled		Y
Min digits	2	N		Identifies the minimum number of values required on entry. ENTER key should not be recognized until minimum is met. Default is 1.		Y
Max digits	2	N		Identifies the maximum number of values allowed to be on entered. Any attempt to enter additional should generate an ERROR tone. Default is 1.		Y

This information is confidential and proprietary of First Data Corporation.
Reproduction without the expressed written consent of First Data Corporation is prohibited.

Dynamic Card Table
PROMPT table

Field Name	Max Field Length	Field Type	Values	Description	Key Column	IN Load File
PED Prompt	1	AN		Identifies the prompt usage on indoor PIN Entry Device. Default is 0. **Value** **Description** 0 No display 1 Display only 2 Enter/display		Y
Pump Prompt	1	AN		Identifies the prompt usage at Pump. Default is 0. **Value** **Description** 0 No display 1 Display only 2 Enter/display		Y
Console Prompt	1	AN		Identifies the prompt usage at cashier's console. Default is 0. **Value** **Description** 0 No display 1 Display only 2 Enter/display		Y
Masking Indicator	1	AN		Specifies if customer input data should be masked when entered. Default is 1. **Value** **Description** 0 Do not mask, display data in clear 1 Mask each byte with an "*" as entered		Y

This information is confidential and proprietary of First Data Corporation.
Reproduction without the expressed written consent of First Data Corporation is prohibited.

Dynamic Card Table
PROMPT table

Field Name	Max Field Length	Field Type	Values	Description	Key Column	IN Load File
Destination Field	1	N	3 or 6 with an embedded decimal point	Specifies the Field ID in the ATL105 Message Format. The input data is to be provided to the host. For destination fields that require an additional data indicator that value will be provided after the input location separated by a decimal point. In particular Data Segment No. 113 (Additional Data Segment) requires a 3 character Additional Data Indicator for every element populated into that field.		Y
Timestamp	12	N		Identifies the CCYYMMDDHHMM of the last update to this entry.		

As an example, assume the following content on this table:

Description: PROMPT FOR ODOMETER
Corporate ID: FIRSTDATA
Version: 00006
PROMPT Code: FM
Display Line 1: ENTER ODOMETER
Display Line 2:
Data Type: N
Min digits: 00
Max digits: 07
PED Prompt: 2

This information is confidential and proprietary of First Data Corporation.
Reproduction without the expressed written consent of First Data Corporation is prohibited.

Pump Prompt:	2
Console Prompt:	2
Masking Indicator:	0
Destination Field:	64
Timestamp:	201201302350

The data sent to the POS/Box would appear as follows:

FM,ENTER ODOMETER,,N,00,07,2,2,2,0,64\

As an example of a field that requires the additional data indicator, assume the following content on this table:

Description:	CVV DATA
Corporate ID:	FIRSTDATA
Version:	00006
PROMPT Code:	CI
Display Line 1:	ENTER CARD ID #
Display Line 2:	
Data Type:	#
Min digits:	04
Max digits:	04
PED Prompt:	2
Pump Prompt:	0
Console Prompt:	0
Masking Indicator:	0
Destination Field:	113.004
Timestamp:	201201302350

The data sent to the POS/Box would appear as follows:

CI,ENTER CARD ID #,,N,04,04,2,0,0,0,113.004\

This information is confidential and proprietary of First Data Corporation.
Reproduction without the expressed written consent of First Data Corporation is prohibited.

2.3 Data Format in Proprietary Data Load Request

The ATL105 Message Format Specification details the Proprietary Data Load request messages. The format defined in the latest version of the ATL105 Message Format Specification is used to request a Card Table Load when instructed by the BUYPASS host. For details, please refer to the latest version of the ATL105 Message Format Specification.

In the latest version of the ATL105 Message Format Specification, Element No. 175 is the Card Table Data. This field has a maximum size of 3700 bytes. The length of this field varies based on the specific block of data being exchanged and is determined by Element No. 84 (Segment Length). The actual content of this data is drawn from the records of the individual card tables with each field separated by a "," (comma) and records separated by a "\" (backslash). Within an individual response message data, multiple tables will not be included. Element Number 174 Card Table Type, identifies the current information content. For example, if the Card Table Type is a 1, which is defined in the specification as BIN Table, then the Card Data block only contains BIN data records. Additionally, records will not be split across response messages, so any Card Data Block should end with a "\" (backslash) character. The content of a record will be the data as indicated in Section 2.2, Card Tables, with the exclusion of the DESCRIPTION column. As an example, the records being sent for the PRODUCT table would consist of Product Code, Display Text 1, and Display Text 2. If a record in the database contained Product Code 001, Display Text 1 of "SUPREMEbbbbbbbb", and a Display Text 2 of "SUPERbbbbbbbbbb" (where 'b' is a blank), the data block would contain 1,SUPREME,SUPER\\.

2.4 Host Databases for Managing the Process

The following sections describe the content and process for the host databases for managing Dynamic Card Tables with the POS Devices.

2.4.1 Card Table Master

The Card Table Master database defines all valid Card Table Identifiers or Versions. A Card Table is defined as an accumulated set of the 6 supporting Card Tables into a Card Table Master Version. For example, Card Table Master version 1 would be defined as a specific version of each of the 6 supporting tables – BIN, RULES, RESTRICTIONS, SAF, PROMPT, and PRODUCT.

A record must exist in this database for any Card Table that is to be downloaded to a POS Device and tracked. Without this entry the host has no reference to which versions of the supporting tables are to be provided. Additionally as described in Section 2.1.2 (Dynamic Card Table Requirements from User Standpoint), the host has no way to determine if a device is processing an out of date version of a component table.

The following table defines the Card Table Master:

First Data.

This information is confidential and proprietary of First Data Corporation.
Reproduction without the expressed written consent of First Data Corporation is prohibited.

Card Table Master				
Field Name	**Size**	**Field Type**	**Description**	**Note**
CT-CORPORATE-ID	10	AN	Identifies a customer group. As an example FIRSTDATA.	This value would be unique to any customers using Dynamic Card Table. This would be used to differentiate Customer FIRSTDATA's Card Table 1 from Customer BUYPASS's Card Table 1.
CT-ID	5	N	Identifies the Card Table Master ID.	The Corporate ID and Card table ID are the links from the Merchant Card Table to the Card Table Master. This subsequently identifies the correct version of the 6 component tables
CT-TIMESTAMP	18	N	Identifies the Julian-timestamp of the last update to this entry.	
CT-BIN-VER	5	N	Identifies the current version of the BIN table for this Card Table Master ID.	
CT-RULES-VER	5	N	Identifies the current version of the RULES table for this Card Table Master ID.	
CT-RESTRICTIONS-VER	5	N	Identifies the current version of the RESTRICTIONS table for this Card Table Master ID.	
CT-SAF-VER	5	N	Identifies the current version of the SAF table for this Card Table Master ID.	
CT-PROMPT-VER	5	N	Identifies the current version of the PROMPT table for this Card Table Master ID.	

First Data.

This information is confidential and proprietary of First Data Corporation.
Reproduction without the expressed written consent of First Data Corporation is prohibited.

Card Table Master				
Field Name	**Size**	**Field Type**	**Description**	**Note**
CT-PRODUCT-VER	5	N	Identifies the current version of the PRODUCT table for this Card Table Master ID.	
CT-MASTER-PREVIOUS	30	N	Identifies the 6 component table Id's of the previous version of this entry.	Supports ability to re-establish the prior working version in the event of an unforeseen issue arose from a new version of the Card Table

Note: The CT-MASTER-PREVIOUS is included as a historical record as well as providing a rollback marker. If one or more of the 6 component table versions is updated, the prior settings are saved to this field. If there was a subsequent problem with the new configuration, the system could be reset to the prior version based on this value. For example, the Master record currently indicates Bin 1, Rules 4, Restrictions 3, SAF 3, Prompt 4 and Product 4. An update is made to increase the SAF to 7 and the PRODUCT to 9. This value would be set to 000010000400003000030000400004. If the updates to SAF and/or PRODUCT resulted in a problem, all affected locations could be moved back to the prior working version since the correct version is known.

First Data.

This information is confidential and proprietary of First Data Corporation.
Reproduction without the expressed written consent of First Data Corporation is prohibited.

2.4.2 Merchant Card Table

The Merchant Card Table database defines the Card Table version to be used by a specific location. BUYPASS host uses this record during Totals processing and Card Table load processing to determine the Card Table version and the component table versions. This in turn facilitates recognizing updated Card Tables that are to be sent to a device.

An entry must exist in this table in order to support Dynamic Card Table functionality. In order to simplify the setup and management, a Global record is supported. This allows a customer to create a single entry that applies to all locations. In the event a location requires a unique set of parameters that can also be defined by creating an entry specific to that location. After reviewing the contents and description of the database fields, this concept will be expanded.

The following table defines the Merchant Card Table:

Field Name	Size	Field Type	Description	Note
MCT-CORPORATE-ID	10	AN	Identifies a customer group. For example FIRSTDATA.	This value would be retrieved from BANKODFI based on Bank FIID – via the SHRSEG file in the BUYPASS host. This is to support the concept of Global records.
MCT-MERCHANT	18	N	Identifies the BUYPASS Merchant ID.	Value of 0 for a Global Record.
MCT-TERM	3	N	Identifies the BUYPASS Terminal ID.	Value of 0 for a Global Record or value of 0 if all POS Devices at this location are to use same Card Table Master Version.
MCT-MASTER-ID	5	N	Identifies the Card Table Master entry that this location uses. This provides the version information for the supporting Card Table files.	
MCT-TIMESTAMP	18	N	Identifies the Julian-timestamp of the last Card Table Load for this location.	
MCT-PREV-MASTER	5	N	Identifies the previous Card Table entry this location used.	Provides a roll back point if a problem is encountered on an upgrade to a new Card Table Master entry.

First Data.

This information is confidential and proprietary of First Data Corporation.
Reproduction without the expressed written consent of First Data Corporation is prohibited.

The concept of a Global record eliminates the need to configure an entry for every location of a customer. For example, FIRSTDATA might have 5000 locations and all the locations are using the same version of the 6 card table components. Instead of loading 5000 entries to this table, a single entry of Corporate ID FIRSTDATA would be configured using Merchant/Term 0 and Card Table Master ID 1. In the future if FIRSTDATA creates Card Table Master 2 and deploy it to Merchant 25, then a single entry would be added to this database. That entry would still be Corporate ID FIRSTDATA but Merchant 25 and Terminal 0 (all POS devices use same processing characteristics). That location would then be configured to the new Card Table on its next Totals request. Once the new Card Table Master is validated in Production, the Global entry record would be updated to Master ID 2 and the Previous Master ID value would be updated to 1. This would result in the remaining 4999 locations getting a Card Table load to the new version on their next Totals request.

There would be no requirement to use a Global record but it would be available if desired. Another use of a Global record could be as a default for locations that get added to the system but for some reason they were never added to this database prior to installation. That would prevent a failed installation and allow the device to be operational under a base set of processing characteristics approved by the customer.

2.4.3 BANKODFI

The BANKODFI database is an existing BUYPASS Batch Database. This database is used to define valid Bank FIIDs and certain data related to the processing for Merchants defined to use a particular FIID. As part of the new functionality, a new field will be added to this database that will be the BANK CORPORATE ID. This field will also be propagated into the SHRSEG file which contains Bank ODFI information that is referenced by the BUYPASS host.

The BANK CORPORATE ID will provide a high level identifier to group Bank FIIDs. This will be used as the Corporate ID key for the various Card Table databases. It will identify the user of a particular set of entries. This will also be used by the process that will extract the table records into the data files to be sent to a POS device during a Dynamic Card Table load request.

Inclusion of this field in the SHRSEG file will facilitate future development for transaction edits. Currently if there are certain customer specific transaction edits that must occur, the application uses the Bank FIID to identify the customer. This means that as Bank FIIDs are added, the application must also be updated. In some cases a customer group could have 10 or more Bank FIIDs which complicates the process of keeping these checks current. By moving to a single consistent identifier the application would not be impacted by future additions of new Bank FIID's. The host would simply use the Corporate Identifier in place of Bank FIID checks.



This information is confidential and proprietary of First Data Corporation.
Reproduction without the expressed written consent of First Data Corporation is prohibited.

2.5 User Interface

2.5.1 Reports

There are no specific report requirements for this functionality. Since the Card Table data is managed in SQL databases, the ability to create on demand SQL ReportWriter reports is available. The use of ReportWriter is sufficient to support this feature.

The Card Master Table and the Merchant Card Table are ENSCRIBE databases. These databases reside on the BUYPASS host and support reporting via the Tandem ENFORM utility.

2.5.2 Screens

To support this functionality, 2 new Requester/Server screens are added and 1 existing screen has been impacted.

The BANKODFI screens defined via the ABNK Requester and Server will be updated to support the new Corporate ID field. This field will be defined as 10 alphanumeric characters. An edit file to be read in should be created to reflect the valid Corporate ID values that may be used. This will ensure that multiple Bank FIIDs for the same customer uses the same valid value. Initially this file should allow spaces to support existing FIID entries.

The new Requester/Server processes will be created to support the Card Table Master database and the Merchant Card Table database. When adding entries to these 2 new databases, the new servers should use the BANKODFI database to verify that a valid Corporate ID is used.

2.5.3 Transaction Terminals

The POS impacts for this functionality are defined via the ATL105 Message Format Specification. This feature is part of the Proprietary Data Load message and is the Dynamic Card Table support function. The data is being provided from the BUYPASS host but it is up to the POS/Box to manage the parameters and apply them to transaction requests.

2.6 Database

Databases required to support this functionality are described in section 2.4 (Host Databases for Managing the Process). All of these databases are standard SQL databases that reside on the BUYPASS Batch node.

2.7 Recovery

This new feature has as its core the new SQL databases. These databases define the following:

- Card Table parameters
- Card Table combinations that are valid

This information is confidential and proprietary of First Data Corporation.
Reproduction without the expressed written consent of First Data Corporation is prohibited.

- Card Table that a particular location is using for processing

All of these databases are standard Tandem TMF Audited databases. These databases provide any recovery requirements that are needed to support the dynamic card table functionality.

2.8 Security

This functionality uses the standard Tandem security configurations currently in use for other modules. Data volumes and files will be secured to limit access to authorized personnel. Existing BUYPASS Standards and Procedures exist to control how and when code changes are installed and who is authorized to do code changes. Operational security is provided by creating the rules for selecting and storing passwords, signing on/off, determining users who are allowed to access different files and/or execute specific transactions.

2.9 Diagnostics

There are no special requirements for new Diagnostics related to this feature.

2.10 Performance

Download of Dynamic Card Data to devices is a background function. This is necessary to eliminate any impact to transaction processing at the POS site. Additionally these requests are processed by supporting applications on the BUYPASS host. This is done to eliminate impact from OLTP at the host by moving the processing out of the normal message flow.

First Data.

This information is confidential and proprietary of First Data Corporation.
Reproduction without the expressed written consent of First Data Corporation is prohibited.

3 Environmental Architecture

This chapter discusses the dynamic card table impacts on the following two platforms:

- Hardware Platform
- Software Platform

3.1 Hardware Platform

The impacts of this functionality on Hardware will be limited to disk space for new databases. There is no new hardware requirements needed to implement this functionality.

3.2 Software Platform

There is no need for any new or updated Operating System software or Third Party supporting applications for this functionality.

This information is confidential and proprietary of First Data Corporation.
Reproduction without the expressed written consent of First Data Corporation is prohibited.

4 Departmental Impacts

This chapter discusses the dynamic card table impacts on the following departments:

- QA/Certification
- Operations
- Production Support
- Communications
- Technical Services
- Billing/Balancing
- Customer Service
- Data Entry
- Marketing Support
- Depot/Maintenance
- Accounting

This information is confidential and proprietary of First Data Corporation.
Reproduction without the expressed written consent of First Data Corporation is prohibited.

4.1 QA/Certification

This document provides additional supporting detail for the Dynamic Card Table functionality. Please contact a BUYPASS representative for additional information.

4.2 Operations

This document provides additional supporting detail for the Dynamic Card Table functionality. Please contact a BUYPASS representative for additional information.

4.3 Production Support

This document provides additional supporting detail for the Dynamic Card Table functionality. Please contact a BUYPASS representative for additional information.

4.4 Communications

This document provides additional supporting detail for the Dynamic Card Table functionality. Please contact a BUYPASS representative for additional information.

4.5 Technical Services

The Dynamic Card Table feature introduces a series of new Databases that will be deployed on both the BUYPASS Online and Batch nodes. Additionally, some of these Databases will also result in ENCRIBE files being produced from the data in the form of the actual Card Table data files that will be sent to the POS device. There will be some additional requirements for disk storage to support this functionality.

4.6 Billing/Balancing

There are no known impacts to this group as part of this functionality.

4.7 Customer Service

There are no known impacts to this group as part of this functionality.

4.8 Data Entry

The Dynamic Card Table feature introduces a new requirement that customers using the new Corporate ID element should use Bank FIIDS. This data provides the cross reference to the various Card Table, Merchant Card Table, and Card Table Master entries.

The Card Table Master database entry and the Merchant Card Table database entry defines valid card tables for use and the specific location where the card table is to be used.

This information is confidential and proprietary of First Data Corporation.
Reproduction without the expressed written consent of First Data Corporation is prohibited.

4.9 Marketing Support

There are no known impacts to this group as part of this functionality.

4.10 Depot/Maintenance

There are no known impacts to this group as part of this functionality.

4.11 Accounting

There are no known impacts to this group as part of this functionality.

This information is confidential and proprietary of First Data Corporation.
Reproduction without the expressed written consent of First Data Corporation is prohibited.

5 Customer Impacts

This chapter discusses the dynamic card table impacts on the following customers:

- Merchant
- End Processor
- Financial/ Proprietary Institutions
- Authorizers/Issuers

5.1 Merchant

This document provides additional supporting detail for the Dynamic Card Table functionality. For more detail, please refer to latest version of the ATL105 Message Format Specification.

5.2 End Processor

There are no known impacts to this group as part of this functionality.

5.3 Financial/Proprietary Institutions

There are no known impacts to this group as part of this functionality.

5.4 Authorizers/Issuers

There are no known impacts to this group as part of this functionality.

This information is confidential and proprietary of First Data Corporation.
Reproduction without the expressed written consent of First Data Corporation is prohibited.